

Assignment 3: Lists & Iterators

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Part 1: Implement a stack using a java.util.LinkedList

```
----- TESTING LINKED LIST STACK -----
Pushing 100
Pushing 200
Pushing 300
Pushing 400
Pushing 500

Current stack:
500
400
300
200
100

Popping
Removed: 500

Current stack:
400
300
200
100

Popping
Removed: 400

Current stack:
300
200
100

Current stack:
200
100

Popping
Removed: 200

Current stack:
100

The stack is not empty. Size: 1

Emptying the stack...
Removed: 100
The stack is empty. Size: 0

Testing pop and peek methods on empty stack...

Popping...
The stack is already empty!
Removed: null
Peek: null
```

Testing methods with strings...
Pushing 3 names

Current stack:
William
Parson
David

Popping twice...
Removed: William
Removed: Parson

Peek: David

Current stack:
David

The stack is not empty. Size: 1

Part 2: Implement a stack using a java.util.ArrayList

----- TESTING ARRAY LIST STACK -----

Pushing 100
Pushing 200
Pushing 300
Pushing 400
Pushing 500

Current stack:
500
400
300
200
100

Popping
Removed: 500

Current stack:
400
300
200
100

Popping
Removed: 400

Current stack:
300
200
100

Popping
Removed: 300

Current stack:
200
100

Popping
Removed: 200

Current stack:
100

The stack is not empty. Size: 1

Emptying the stack...
Removed: 100
The stack is empty. Size: 0

Testing pop and peek methods on empty stack...

Popping...
The stack was already empty!
Removed: null
The stack is already empty!
Peek: null

Testing methods with strings...
Pushing 3 names

Current stack:
William
Parson
David

Popping twice...
Removed: William
Removed: Parson

Peek: David

Current stack:
David

The stack is not empty. Size: 1

Part 3: Implement a queue using a java.util.LinkedList

```
----- TESTING LINKED LIST QUEUE -----  
Adding 100 to rear  
Adding 200 to front  
Adding 300 to rear  
Adding 400 to front  
Adding 500 to rear  
Adding 600 to front  
  
Current queue:  
600  
400  
200  
100  
300  
500  
  
Dequeuing rear  
Removed: 500  
  
Current queue:  
600  
400  
200  
100  
300  
  
Dequeuing front  
Removed: 600  
  
Current queue:  
400  
200  
100  
300  
  
Dequeuing rear  
Removed: 300  
  
Current queue:  
400  
200  
100  
  
Dequeuing front  
Removed: 400  
  
Current queue:  
200  
100  
  
The queue is not empty. Size: 2  
  
Emptying the queue...  
  
Dequeuing rear  
Removed: 100  
  
Dequeuing front  
Removed: 200  
The queue is empty. Size: 0
```

```
Testing dequeue, first, and last methods on empty stack...
```

```
Dequeuing rear
The queue was already empty!
Removed: null
Dequeuing front
The queue was already empty!
Removed: null
First: null
Last: null
```

```
Testing methods with strings...
Enqueueing 3 names
```

```
Current queue:
Parson
David
William
```

```
Dequeuing rear
Removed: William
```

```
Dequeuing front
Removed: Parson
```

```
First: David
Last: David
```

```
Current queue:
David
```

```
The queue is not empty. Size: 1
```

Part 4: Implement a queue using a `java.util.ArrayList`

```
----- TESTING ARRAY LIST QUEUE -----
```

```
Adding 100 to rear
Adding 200 to front
Adding 300 to rear
Adding 400 to front
Adding 500 to rear
Adding 600 to front
```

```
Current queue:
```

```
600
400
200
100
300
500
```

Dequeueing rear
Removed: 500

Current queue:
600
400
200
100
300

Dequeueing front
Removed: 600

Current queue:
400
200
100
300

Dequeueing rear
Removed: 300

Current queue:
400
200
100

Dequeueing front
Removed: 400

Current queue:
200
100

The queue is not empty. Size: 2

Emptying the queue...

Dequeueing rear
Removed: 100

Dequeueing front
Removed: 200
The queue is empty. Size: 0

Testing dequeue, first, and last methods on empty stack...

Dequeueing rear
The queue was already empty!
Removed: null

Dequeueing front
The queue was already empty!
Removed: null
The queue is already empty!

First: null
The queue is already empty!
Last: null

Testing methods with strings...
Enqueueing 3 names

Current queue:
Parson
David
William

Dequeueing rear
Removed: William

Dequeueing front
Removed: Parson

First: David
Last: David

Current queue:
David

The queue is not empty. Size: 1
